A machine-checked two-colour polymer gas on $\mathbb{Z}^4$: encoding, adjacency degrees, the box stage, and the Kotecký–Preiss condition down to Dobrushin

Yoshikazu Hosomura
with the assistance of Claude (Claude Code, Anthropic)

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wvbks0 [at] gmail.com · `https://computoergosum.com/en/index.html`

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Abstract

We describe a Lean 4 formalisation of the combinatorial half of a cluster-expansion argument for a gas of connected clusters of plaquettes on $\mathbb{Z}^4$. A cluster X is encoded as a two-colour set: its *thin* plaquettes and its *defect* links (links of which X contains at least m of the six plaquettes), each defect labelled by $X \cap D_\ell$. Lean proves that the encoding preserves connectedness, that a cluster is recovered from its code and labels, and that the label sum factorises over links. On concrete types (points $\texttt{Fin}\,4 \to \mathbb{Z}$, plaquettes = point $\times\,\texttt{Fin}\,6$, links = point $\times\,\texttt{Fin}\,4$) the kernel decides the two-colour degrees 20, 44, 66, 122 and the 20 box shifts between link-sharing plaquettes (boxes of 16 points, 96 plaquettes). The Kotecký–Preiss per-point sum is then bounded on the plaquette and box stages and carried down to Dobrushin's condition and to $Z > 0$, $\log Z \geq -\sum a$. Activities enter only through hypotheses, a product bound or the activity-sum hypothesis (H-sum) $\sum_{X \ni x} |z(X)| e^{2|X|} \leq 1$; no value or estimate of any activity is given. An identity rewrites the per-box sum of the box gas as a sum over plaquette clusters, so that (H-sum) yields Dobrushin's condition and the bounds on Z for the box gas in one composed theorem; a bridge bounds the activity of the clusters meeting a box by plaquette-stage sums. Thirteen files contain 518 theorem declarations, 163 distinct statements after removing copies; 16 are finite counts closed by `decide +kernel`, the rest are proofs.

1 Introduction

Cluster expansions for lattice gauge theories at weak coupling separate the configuration into a region where every plaquette variable is small and a set of *large-field* plaquettes, and then treat the connected components of the large-field set as polymers of a gas whose activities are small (for the setting of the Yang–Mills problem see [2]). The analytic work lies in bounding the activities. A second, purely combinatorial, half is needed as well: one has to count connected clusters on the lattice with enough precision that a smallness condition on a single plaquette (or a single block) turns into the Kotecký–Preiss condition [3] for the whole gas, and this condition in turn into Dobrushin's condition [1] and lower bounds on the partition function. This note is about that combinatorial half, written in Lean 4 against Mathlib [4], for one concrete organisation of the clusters that arose in the authors' work on a lattice $SU(2)$ model in four dimensions. Background on that programme is collected at `https://computoergosum.com/en/principia/yang-mills/index.html`; nothing below depends on it.

The organisation is *two-colour*. Plaquettes that sit around a common link are strongly correlated, so a cluster that contains many of the six plaquettes of one link is better counted as a single object, a *defect link*, than as many independent plaquettes. A cluster is therefore encoded as a set of thin plaquettes together with a set of defect links, each defect link labelled by the subset of its six plaquettes that the cluster contains. The KP sum is then a sum over connected sets of a two-colour graph with colour-dependent weights and degrees. The note records what is formalised (§§2–7), how the lattice objects are represented (§5), which statements are decided by the kernel and which are proved (§8), and which inputs remain assumptions (§9). The authors' working records (unpublished) contain the numerical exploration that motivated the choices; none of it is used here.

Status of the results. Every implication stated as a theorem below is a Lean 4 theorem that passes its check (no `sorry`, no `axiom` declaration, no `native_decide`; axioms `propext`, `Classical.choice`, `Quot.sound`). The finite counts on $\mathbb{Z}^4$ are decided by `decide +kernel`. The activities of a gauge model are *not* treated: the bound on the activities is a hypothesis of every theorem, either in product form (H-2col) or in summed form (H-sum) (§2). That the large-field clusters of a gauge model form a polymer gas of the kind considered here, with compatibility given by disjoint supports, is an assumption stated in §9 and is not formalised. Where we say "proved" we mean "proved in Lean"; the rest of the text is explanation.

2 The setting: clusters, activities, and the two conditions

Graphs and connected sets. Let P be a type with decidable equality and adj a relation on P (symmetry is never used). For a finite $X \subseteq P$, Conn_X is the reflexive–transitive closure of $\{(a,b) : a,b \in X,\ \mathrm{adj}(a,b)\}$; X is *connected* if $\mathrm{Conn}_X(a,b)$ for all $a,b \in X$. A finite C is *rooted* at R if $R \subseteq C$ and every $c \in C$ is reached from some $\rho \in R$ inside C. For weights $x : P \to [0,\infty)$ put

$$W_g(U,R) = \sum_{C \subseteq U,\ C \text{ rooted at } R} \ \prod_{c \in C} x(c)$$

(`Wg`).

The gas. A *cluster* is a finite connected set X of plaquettes (adjacent = distinct and sharing a link) inside a finite region Λ_F; $\mathcal{X}$ is a finite family of clusters, and $z : \mathcal{X} \to \mathbb{R}$ is the activity. Two clusters are incompatible when their supports meet. With supp the support map (the identity on the plaquette stage, the set of boxes on the box stage, §6) and $\mu(X) = |z(X)| e^{2|\operatorname{supp} X|}$ we use:

- the *Kotecký–Preiss per-point condition* (the hypothesis `hsite` of `KPToDobrushin`):

$$\forall x : \quad \sum_{X \in \mathcal{X},\ x \in \operatorname{supp} X} |z(X)|\, e^{2|\operatorname{supp} X|} \ \le\ 1; \tag{KP}$$

- *Dobrushin's condition* with $a(X) = \log(1+\mu(X))$:

$$|z(X)| \ \le\ (1 - e^{-a(X)}) \exp\Big(- \sum_{X' \neq X,\ \operatorname{supp} X' \cap \operatorname{supp} X \neq \emptyset} a(X') \Big). \tag{D}$$

The two activity hypotheses. Fix a set Λ_L of links and a threshold $m \ge 1$. For a link ℓ let D_ℓ be the set of its six plaquettes. For a cluster X,

$$\mathrm{def}(X) = \{\ell \in \Lambda_L :\ |X \cap D_\ell| \ge m\}, \qquad \mathrm{thin}(X) = \{p \in X :\ p \notin D_\ell \text{ for all } \ell \in \mathrm{def}(X)\}.$$

- **(H-2col)** (product form; the hypothesis `hz`): $|z(X)| \le \varepsilon_0^{|\operatorname{thin} X|} \prod_{\ell \in \operatorname{def} X} \eta(\ell, X \cap D_\ell)$ for numbers $\varepsilon_0 \ge 0$ and $\eta \ge 0$, together with a bound on the *label sum* of each link (the hypothesis `hlink`), $\sum_{S \subseteq D_\ell} \eta(\ell, S) e^{2|S|} \le c_L$ on the plaquette stage, $\sum_{S \subseteq D_\ell} \eta(\ell, S) \le \varepsilon_L$ on the box stage.

- **(H-sum)** (summed form): the inequality (KP) itself, assumed for the gas, either with $\operatorname{supp} = \mathrm{id}$ (plaquette stage) or with supp the box support of §6 (box stage).

(H-2col) is the form needed to *derive* (KP) from per-object smallness; (H-sum) is the form in which (KP) is taken as the assumption and only the passage to the box gas and to (D) is formalised. Where a theorem quoted below contains numerical constants in its hypotheses (for instance $\varepsilon_0 \le 2.5 \cdot 10^{-3}$), these are the thresholds of a sufficient condition proved in Lean; they are not estimates of any activity.

3 Two-colour connected sets and the per-point sum

The counting rests on a weighted form of the classical tree-graph/generating-function bound for connected sets.

Lemma 3.1 (Lean: `Wg_le`). *Let* $\mathrm{nb}(p)$ *be finite sets with* $p \ne q \wedge \mathrm{adj}(p,q) \Rightarrow q \in \mathrm{nb}(p)$*, and let* $x, \tau \ge 0$ *satisfy* $x(p) \prod_{q \in \mathrm{nb}(p)} (1 + \tau(q)) \le \tau(p)$ *for all* p*. Then* $W_g(U, R) \le \prod_{r \in R} \tau(r)$ *for all finite* U, R*.*

The proof is a strong induction on $|U|$: removing one root r replaces it by a subset Q of its neighbours outside R (`rooted_erase`, `Wg_step`). The hypothesis on nb excludes $q = p$; this weakening (relative to the first version of the file) lets a defect link be adjacent to itself in the two-colour graph without entering its own neighbourhood. For a single colour and constant weights the same induction gives the count $\#\{C \ni p \text{ connected}, |C| = n\} \le e^n d^{n-1}$ when every point has at most $d \ge 1$ neighbours (`connected_count_le`), and the one-colour KP sum $\sum_{C \ni p} |z(C)| e^{2|C|} \le 1/d$ under $e^3 d\varepsilon \le 1$ (`point_kp_sum_le`).

Now colour the points, $\mathrm{col} : P \to \{0, 1\}$ (in Lean `Bool`; `false` = thin plaquette, `true` = defect link), and let d_{ab} bound the number of colour-b neighbours of a colour-a point.

Proposition 3.2 (Lean). *Let* $\varepsilon_b, \theta_b \ge 0$ *and* $a_b \in \mathbb{R}$*. If* $|z(C)| \le \prod_{q \in C} \varepsilon_{\mathrm{col}(q)}$ *for every finite* C *and*

$$\varepsilon_b \, e^{a_b} (1 + \theta_0)^{d_{b0}} (1 + \theta_1)^{d_{b1}} \le \theta_b \qquad (b = 0, 1),$$

then for every finite U*, every point* p *and every finite root set* S*, with* C *ranging over the connected subsets of* U*,*

$$\sum_{C \ni p \text{ conn.}} |z(C)| \prod_{q \in C} e^{a_{\mathrm{col}(q)}} \le \theta_{\mathrm{col}(p)}, \qquad \sum_{C \cap S \ne \emptyset \text{ conn.}} |z(C)| \prod_{q \in C} e^{a_{\mathrm{col}(q)}} \le \sum_{r \in S} \theta_{\mathrm{col}(r)}.$$

(`two_color_point_sum`, `two_color_root_set_sum`; the version with point weights $x \le \xi_{\mathrm{col}}$ *in place of* z *is* `two_color_root_gf`*.)*

Lemma 3.3 (Lean). *With* $(d_{00}, d_{01}, d_{10}, d_{11}) = (20, 48, 72, 122)$*,* $(a_0, a_1) = (2, 12)$ *and* $(\theta_0, \theta_1) = (1/20, 10^{-4})$*, the two inequalities of Proposition 3.2 hold whenever* $\varepsilon_0 \le 2.5 \cdot 10^{-3}$ *and* $\varepsilon_1 \le 1.5 \cdot 10^{-11}$*; moreover* $\theta_0 + 4\theta_1 \le 1$ *(`lattice_fix`, `lattice_face_total`).*

The proof uses $e \le 2.7182818286$ (`exp_one_le`) and `norm_num`. The degrees 48 and 72 are upper bounds; the exact degrees 44 and 66 are decided in §5 and enter through "$\le$". The weight $e^{a_1} = e^{12} = (e^2)^6$ is the KP weight of the at most six plaquettes of a defect.

4 The encoding

Let F (plaquettes) and L (links) be types with decidable equality, $D : L \to \mathrm{Finset}\, F$, adj_F a relation on F, Λ_L a finite set of links and $m \geq 1$. The code of X is

$$\mathrm{enc}(X) = \mathrm{thin}(X) \sqcup \mathrm{def}(X) \;\subseteq\; F \oplus L,$$

and the two-colour adjacency `cadj` on $F \oplus L$ is

$$\begin{aligned}
&p \sim q \iff \mathrm{adj}_F(p,q),\\
&p \sim \ell \iff p \notin D_\ell \;\wedge\; \exists q \in D_\ell,\ \mathrm{adj}_F(p,q), \qquad \ell \sim p \iff p \notin D_\ell \;\wedge\; \exists q \in D_\ell,\ \mathrm{adj}_F(q,p),\\
&\ell \sim \ell' \iff \exists p \in D_\ell,\ \exists q \in D_{\ell'},\ p = q \;\vee\; \mathrm{adj}_F(p,q).
\end{aligned}$$

Lemma 4.1 (Lean, `LatticeTwoColorKP`)**.** *For every finite $X \subseteq F$:*

*(i) (*connectedness*, **`enc_connected`**) if X is connected for adj_F and $m \geq 1$, then $\mathrm{enc}(X)$ is connected for **`cadj`**;*

*(ii) (*recovery*, **`decode`**, **`enc_eq`**, **`thin_disjoint`**) $X = \mathrm{thin}(X) \cup \big(X \cap \bigcup_{\ell \in \mathrm{def}\, X} D_\ell\big)$; equal codes have equal thin parts and equal defect sets; and $\mathrm{thin}(X)$ is disjoint from $\bigcup_{\ell \in \mathrm{def}\, X} D_\ell$;*

*(iii) (*size*, **`card_le_enc`**) $|X| \leq |\,\mathrm{thin}\, X| + \sum_{\ell \in \mathrm{def}\, X} |X \cap D_\ell|$;*

*(iv) (*label sum*, **`label_sum_le`**) for $g \geq 0$ and any finite $R \subseteq L$,*

$$\sum_{T \subseteq \bigcup_{\ell \in R} D_\ell} \prod_{\ell \in R} g(\ell, T \cap D_\ell) \;\leq\; \prod_{\ell \in R} \sum_{S \subseteq D_\ell} g(\ell, S),$$

even when the D_ℓ overlap.

The connectedness proof maps each plaquette $f \in X$ to its image $\mathrm{img}(f)$, a defect link containing it (chosen with `Classical.choose`) or f itself (`img_spec`); adjacent plaquettes have adjacent or equal images (`img_adj`), so paths in X map to paths in $\mathrm{enc}(X)$, and every point of $\mathrm{enc}(X)$ is the image of, or adjacent to the image of, a plaquette of X. Injectivity is used in the following form: on the fibre $\{X : \mathrm{enc}(X) = C\}$ all clusters share $\mathrm{thin}(X)$ and $\mathrm{def}(X)$, so by (ii) the map $X \mapsto X \cap \bigcup_{\ell \in \mathrm{def}} D_\ell$ is injective there; this is the step `hinj` inside the proof of the next proposition, combined with (iv).

Proposition 4.2 (Lean: `encoded_sum_le`)**.** *Under (H-2col), for any predicate Q satisfied by all codes $\mathrm{enc}(X)$, $X \in \mathcal{X}$,*

$$\sum_{X \in \mathcal{X}} |z(X)| e^{2|X|} \;\leq \sum_{\substack{C \subseteq \Lambda_F \sqcup \Lambda_L \\ Q(C),\ C \text{ connected}}} \prod_{q \in C} x_w(q),$$

*with connectedness for **`cadj`**, $x_w(p) = \varepsilon_0 e^2$ and $x_w(\ell) = \sum_{S \subseteq D_\ell} \eta(\ell, S) e^{2|S|}$.*

The version with weight $e^{c|X|}$ for any $c \geq 0$ is `encoded_sum_le_c`; it is used with $c = 0$ on the box stage.

5 The lattice $\mathbb{Z}^4$ and the adjacency degrees

Representation. Points, plaquettes and links are

```
abbrev Pt := Fin 4 → ℤ
abbrev Face := Pt × Fin 6
abbrev Link := Pt × Fin 4
```

The plaquette index $k = 0, \dots, 5$ stands for the plane $(0,1),(0,2),(0,3),(1,2),(1,3),(2,3)$ and the first component is the base point. Two tables at the origin, `links0` (the four links of the plaquette $(0,k)$; for the plane (μ,ν) these are $(0,\mu),(e_\mu,\nu),(e_\nu,\mu),(0,\nu)$) and `faces0` (the six plaquettes containing the link $(0,\mu)$, namely $(0,\{\mu,\nu\})$ and $(-e_\nu,\{\mu,\nu\})$ for $\nu \neq \mu$), are translated by the embeddings `tF`, `tL` to give `linksOf` and `facesOf` $= D$. Their consistency, $f \in D_\ell \iff \ell \in \mathrm{linksOf}(f)$, is `mem_facesOf_iff`, reduced by translation to the origin and proved there by case analysis (`key0`). Plaquette adjacency `faceAdj` means: distinct, and both in some D_ℓ.

Neighbourhoods and degrees. The two-colour neighbourhoods are translates of finite sets at the origin (writing ∂f for the links of f and $N_{FF}(f)$ for the translate of N^0_{FF}):

$$N^0_{FF}(k) = \Big(\bigcup_{\ell \in \partial(0,k)} D_\ell \Big) \setminus \{(0,k)\},$$

$$N^0_{FL}(k) = \Big\{ \ell \in \bigcup_{g \in N^0_{FF}(k)} \partial g : \ (0,k) \notin D_\ell \Big\},$$

$$N^0_{LF}(\mu) = \Big(\bigcup_{f \in D_{(0,\mu)}} N_{FF}(f) \Big) \setminus D_{(0,\mu)},$$

$$N^0_{LL}(\mu) = \Big(\bigcup_{f \in E_\mu} \partial f \Big) \setminus \{(0,\mu)\},$$

with $E_\mu = D_{(0,\mu)} \cup \bigcup_{f \in D_{(0,\mu)}} N_{FF}(f)$. Lean shows that every `cadj`-neighbour (other than the point itself) lies in the translated set (`adj_mem_nbFF`, `fl_mem`, `lf_mem`, `ll_mem`, assembled in `lattice_hnb`) and decides the sizes:

```
theorem card_nbFF0 : ∀ k, (nbFF0 k).card = 20 := by decide +kernel
theorem card_nbFL0 : ∀ k, (nbFL0 k).card = 44 := by decide +kernel
theorem card_nbLF0 : ∀ μ, (nbLF0 μ).card = 66 := by decide +kernel
theorem card_nbLL0 : ∀ μ, (nbLL0 μ).card = 122 := by decide +kernel
theorem card_links0 : ∀ k, (links0 k).card = 4 := by decide +kernel
```

So a plaquette has 20 plaquette neighbours and 44 defect-link neighbours, and a defect link has 66 plaquette neighbours and 122 defect-link neighbours; every plaquette has 4 links (`lattice_hfour`: at most 4 links of Λ_L contain a given plaquette). These close the hypotheses `hnb`, `hdeg` (`lattice_hdeg`, against the upper bounds $(20, 48, 72, 122)$ of Lemma 3.3) and `hfour` of the abstract theorem, leaving only the activity hypotheses:

Theorem 5.1 (Lean: `lattice_kp_site`, `lattice_dobrushin`)**.** *On $\mathbb{Z}^4$, under (H-2col) with $\varepsilon_0 \le 2.5 \cdot 10^{-3}$ and label sums $\le 1.5 \cdot 10^{-11} e^{12}$, (KP) holds on the plaquette stage; if moreover $|X| \ge 1$ and $|z(X)| e^{2|X|} \le e - 1$ for all $X \in \mathcal{X}$, then (D) holds for every $X \in \mathcal{X}$.*

```
theorem lattice_kp_site (ΛF : Finset Face) (ΛL : Finset Link) (m : ℕ) (hm : 1 ≤ m)
    (𝒳 : Finset (Finset Face)) (h𝒳 : ∀ X ∈ 𝒳, X ⊆ ΛF ∧ IsConnectedSet faceAdj X)
    (z : Finset Face → ℝ) (ε₀ : ℝ) (hε₀ : 0 ≤ ε₀) (hε₀' : ε₀ ≤ 2.5e-3)
    (η : Link → Finset Face → ℝ) (hη : ∀ ℓ S, 0 ≤ η ℓ S)
    (hlink : ∀ ℓ, ∑ S ∈ (facesOf ℓ).powerset, η ℓ S * Real.exp (2 * S.card) ≤ 1.5e-11
      * Real.exp 12)
    (hz : ∀ X ∈ 𝒳, |z X| ≤ (∏ _p ∈ thin facesOf ΛL m X, ε₀) *
      ∏ ℓ ∈ defects facesOf ΛL m X, η ℓ (X ∩ facesOf ℓ)) :
    ∀ x : Face, ∑ X ∈ 𝒳.filter (fun s => x ∈ id s), |z X| * Real.exp (2 * (id
      X).card) ≤ 1 :=
```

The proof is `encoded_kp_site`: Proposition 4.2 with $Q(C) =$ "C meets the root set of x", where the root set of a plaquette x is $\{x\} \sqcup \{\ell \in \Lambda_L : x \in D_\ell\}$ (`rootSet`, `enc_meets_root`); Proposition 3.2 with Lemma 3.3 bounds the result by $\theta_0 + 4\theta_1 \le 1$.

A wider adjacency. The file `Lattice24KP` repeats the construction with plaquette adjacency enlarged by the four parallel plaquettes across an elementary cube (`cubeOpp0`, `faceAdj24`). The decided degrees are 24, 44, 66, 122 (`card_nbFF24_0`, ..., `card_nbLL24_0`), the arithmetic condition holds for $\varepsilon_0 \le 2.1 \cdot 10^{-3}$, $\varepsilon_1 \le 1.5 \cdot 10^{-11}$ with $\theta = (1/23, 10^{-4})$ (`lattice24_fix`), and the analogue of Theorem 5.1 is `lattice_kp_site24`. At $\varepsilon_0 \ge 2.5 \cdot 10^{-3}$ no $\theta \ge 0$ satisfies the thin inequality with degrees $(24, 48)$ (`face24_fails`).

6 The box stage

Boxes and shifts. Let $\mathrm{blk}(x) = (\lfloor x_i/2 \rfloor)_i$ (`blk`, integer division). The box of $b \in \mathbb{Z}^4$ is $\{x : \mathrm{blk}(x) = b\} = 2b + \{0,1\}^4$, with 16 points (`card_corners`), 96 plaquettes based in it (`boxFaces`, `card_boxFaces`: ≤ 96) and 64 links (`card_boxLinks`: ≤ 64). A plaquette belongs to the box of its base point. The set of box shifts between plaquettes that share a link is computed at the origin, over the 16 positions of a base point in its box, and decided:

```
theorem card_boxOff : (boxOff.erase 0).card = 20 := by decide +kernel
```

`box_offset` shows that every link-sharing pair of plaquettes has its box shift in `boxOff`; hence the box adjacency has degree ≤ 20 (`box_degree`) and the box image of a connected cluster is connected (`box_image_connected`). The 20 shifts are not the 8 nearest-neighbour shifts alone: `box_diag` exhibits two plaquettes sharing the link $((2,0,0,0),2)$ whose boxes differ by $(1,-1,0,0)$. (Enumerating `boxOff` outside Lean gives the 8 shifts $\pm e_\mu$ and the 12 shifts $e_\mu - e_\nu$; only the total 20 is decided in Lean.)

The two-colour box gas. Points of the box gas are pairs (box, colour). Thin–thin shifts are the 20 above (`card_offFF`); for pairs involving a defect, Lean decides that all point shifts in $N^0_{FL}, N^0_{LF}, N^0_{LL}$ have coordinates in $[-2,2]$ (`nbFL0_bdd`, `nbLF0_bdd`, `nbLL0_bdd`), hence box shifts in $\{-1,0,1\}^4$ (`blk_sub_mem_cube`, `card_cube81`), and uses the degrees $(20, 81, 81, 80)$ (`card_offc`, `box_hdeg`). These are upper bounds; the exact values $39, 39, 56$ found by enumeration outside Lean are not used. The box image bimg of a `cadj`-connected set is connected for the box adjacency (`bimg_connected`). With the product hypothesis $|z_B(\Gamma)| \le \prod_{q \in \Gamma} \varepsilon_{\mathrm{col}(q)}$ over boxes, the arithmetic condition holds for $\varepsilon_{\mathrm{thin}} \le 2.55 \cdot 10^{-3}$, $\varepsilon_{\mathrm{def}} \le 10^{-9}$, $\theta = (1/20, 10^{-6})$ (`box_fix`), giving (KP) with $\mathrm{supp}\,\Gamma$ the set of boxes (`box_kp_site`, `box_hsite`) and (D) (`box_dobrushin`); `box_fails` shows that at $\varepsilon_{\mathrm{thin}} \ge 3.068 \cdot 10^{-3}$ no $\theta \ge 0$ satisfies the thin inequality.

From the plaquette stage to the box gas. Let $\mathrm{boxImage}(X) = \mathrm{bimg}(\mathrm{enc}\, X)$ and define the box activity as the sum over the fibre,

$$z_B(\Gamma) = \sum_{X \in \mathcal{X},\ \mathrm{boxImage}(X) = \Gamma} |z(X)|$$

(`zBox`). Under (H-2col) with $\sum_{S \subset D_\ell} \eta(\ell, S) \le \varepsilon_L$, the fibre sum is bounded by a product over boxes with $\varepsilon_{\mathrm{box}}(\mathrm{thin}) = (1+\varepsilon_0)^{96} - 1$ and $\varepsilon_{\mathrm{box}}(\mathrm{def}) = (1+\varepsilon_L)^{64} - 1$ (`fiber_box_le`, `cell_prod_le`, `box_hz_from_face`), and `face_to_box_dobrushin` carries this to (D) for the box gas when $\varepsilon_0 \le 2.6 \cdot 10^{-5}$, $\varepsilon_L \le 1.5 \cdot 10^{-11}$ (`epsBox_numbers`). This route needs the product form over boxes; we keep it as a formal statement but do not rely on it (§9).

The route used instead takes (H-sum) as the assumption. The box-stage sum of the box gas is *equal* to a sum over plaquette clusters:

Theorem 6.1 (Lean: `box_hsite_eq_face_sum`, `box_hsite_of_face_sum`)**.** *For every box b,*

$$\sum_{\Gamma\in\mathrm{boxImage}(\mathcal{X}),\ b\in\operatorname{supp}\Gamma} |z_B(\Gamma)|\, e^{2|\operatorname{supp}\Gamma|} = \sum_{X\in\mathcal{X},\ b\in\operatorname{supp}\mathrm{boxImage}(X)} |z(X)|\, e^{2|\operatorname{supp}\mathrm{boxImage}(X)|},$$

where $\operatorname{supp}\Gamma$ *is the set of boxes of* Γ*. Consequently (H-sum) in the box form, "the right-hand side is* ≤ 1 *for every* b*", is exactly (KP) for the box gas, with no product hypothesis.*

```
theorem box_hsite_eq_face_sum (ΛL : Finset Link) (m : ℕ) (𝒳 : Finset (Finset Face))
      (z : Finset Face → ℝ)
    (b : Pt) :
    ∑ Γ ∈ (𝒳.image (boxImage ΛL m)).filter (fun s => b ∈ boxSupp s),
        |zBox ΛL m 𝒳 z Γ| * Real.exp (2 * (boxSupp Γ).card) =
      ∑ X ∈ 𝒳.filter (fun X => b ∈ boxSupp (boxImage ΛL m X)),
        |z X| * Real.exp (2 * (boxSupp (boxImage ΛL m X)).card) := by
```

The proof uses only $|z_B| = z_B$ (a sum of absolute values) and regrouping by fibres (`sum_fiberwise_of_maps_to`). Feeding `box_hsite_of_face_sum` into `dobrushin_cond_of_kp` with supp = `boxSupp` gives (D) for the box gas, and Theorem 7.2 then gives the partition function; this composition is stated as one theorem, Theorem 7.3 in §7.

The bridge. The second use of the plaquette stage is to bound, box by box, the total activity of the clusters that touch a box. The root set of a box b is its 96 plaquettes together with the links of Λ_L having a plaquette in the box, at most $96 \cdot 4 = 384$ of them (`boxRoot`, `card_boxRoot_links`); the code of a cluster that meets the box meets this root set (`enc_meets_boxRoot`).

Theorem 6.2 (Lean: `box_event_bridge`)**.** *Under (H-2col) without KP weights, with label sums* $\sum_{S\subseteq D_\ell}\eta(\ell,S) \le \varepsilon_L$*, and if* $\theta_0,\theta_1 \ge 0$ *satisfy* $\varepsilon_0(1+\theta_0)^{20}(1+\theta_1)^{48} \le \theta_0$ *and* $\varepsilon_L(1+\theta_0)^{72}(1+\theta_1)^{122} \le \theta_1$*, then for every box* b

$$\sum_{X\in\mathcal{X},\ X\cap\mathrm{boxFaces}(b)\neq\emptyset} |z(X)| \ \le\ 96\,\theta_0 + 384\,\theta_1.$$

```
theorem box_event_bridge (ΛF : Finset Face) (ΛL : Finset Link) (m : ℕ) (hm : 1 ≤ m)
    (𝒳 : Finset (Finset Face)) (h𝒳 : ∀ X ∈ 𝒳, X ⊆ ΛF ∧ IsConnectedSet faceAdj X)
    (z : Finset Face → ℝ) (εs εL : ℝ) (hεs : 0 ≤ εs) (η : Link → Finset Face → ℝ) (hη
      : ∀ ℓ S, 0 ≤ η ℓ S)
    (hlink : ∀ ℓ, ∑ S ∈ (facesOf ℓ).powerset, η ℓ S ≤ εL)
    (hz : ∀ X ∈ 𝒳, |z X| ≤ (∏ _p ∈ thin facesOf ΛL m X, εs) *
      ∏ ℓ ∈ defects facesOf ΛL m X, η ℓ (X ∩ facesOf ℓ))
    (θ : Bool → ℝ) (hθ : ∀ c, 0 ≤ θ c)
    (hfix : ∀ a, (fun c => if c then εL else εs) a * (1 + θ false) ^ latticeDeg a
      false *
      (1 + θ true) ^ latticeDeg a true ≤ θ a) (b : Pt) :
    ∑ X ∈ 𝒳.filter (fun X => (X ∩ boxFaces b).Nonempty), |z X| ≤ 96 * θ false + 384 *
      θ true := by
```

The proof is `encoded_sum_le_c` with $c = 0$ and $Q(C)$ = "C meets the root set of b", followed by `two_color_root_gf` on $\mathbb{Z}^4$. In words: the total activity of the clusters that meet box b is controlled by the two-colour sums of the plaquette stage, without passing through a

product over boxes. Turning this into a bound on the probability that a box contains a large plaquette requires a Peierls-type inequality $\mu(\text{box } b \text{ bad}) \le \sum_{X \cap b \neq \emptyset} \rho^*(X)$ for the measure, which is not formalised (§9). The file also contains two arithmetic instances of the hypotheses of Theorem 6.2 with literal inputs (`bridge_numbers_gamma`, `bridge_numbers_chernoff`); their inputs are not discussed in this note.

7 From the Kotecký–Preiss sum to Dobrushin and to the partition function

Theorem 7.1 (Lean: `KPToDobrushin.dobrushin_cond_of_kp`)**.** *Let* $\mathrm{supp} : P \to \mathrm{Finset}\, S$*,* U *finite,* $|\operatorname{supp}\gamma| \ge 1$*, (KP) for* U*, and* $|z(\gamma)|e^{2|\operatorname{supp}\gamma|} \le e - 1$ *for* $\gamma \in U$*. Then (D) holds for every* $\gamma \in U$*.*

```
theorem dobrushin_cond_of_kp (supp : P → Finset S) (z : P → ℝ) (U : Finset P)
    (hsize : ∀ γ ∈ U, 1 ≤ (supp γ).card)
    (hsite : ∀ x, ∑ γ' ∈ U.filter (fun s => x ∈ supp s),
      |z γ'| * Real.exp (2 * (supp γ').card) ≤ 1)
    (hsmall : ∀ γ ∈ U, |z γ| * Real.exp (2 * (supp γ).card) ≤ Real.exp 1 - 1) :
    ∀ γ ∈ U, |z γ| ≤
      (1 - Real.exp (-Real.log (1 + |z γ| * Real.exp (2 * (supp γ).card)))) *
        Real.exp (-∑ γ' ∈ (U.erase γ).filter (fun s => (supp γ ∩ supp s).Nonempty),
          Real.log (1 + |z γ'| * Real.exp (2 * (supp γ').card))) := by
```

The proof bounds $a \le \mu$ (from $\log(1+t) \le t$), the sum over polymers meeting $\operatorname{supp}\gamma$ by the sum over points of $\operatorname{supp}\gamma$ (`intersect_sum_le`), hence by $|\operatorname{supp}\gamma|$ by (KP); the remaining inequality $|z| \le (1 - e^{-a})e^{-n}$ is closed using $|z|e^{2n} \le e - 1$ and $n \ge 1$.

Theorem 7.2 (Lean: `PolymerGasDobrushin.dobrushin`, `Z_ge_exp`, `log_Z_ge`)**.** *For a finite gas with symmetric incompatibility relation, real activities of any sign, and* $Z(\Gamma) = \sum_{t \subseteq \Gamma \text{ compatible}} \prod_{s \in t} z(s)$*: if* $a \ge 0$ *is any function and (D) holds with this* a *in place of* $\log(1+\mu)$ *and "incompatible" in place of "supports meet", then for every* $\Gamma \subseteq U$*,* $Z(\Gamma) > 0$*,* $e^{-a(\gamma)} Z(\Gamma \setminus \gamma) \le Z(\Gamma)$ *for* $\gamma \in \Gamma$*,* $Z(\Gamma) \ge \exp(-\sum_{\gamma \in \Gamma} a(\gamma))$ *and* $\log Z(\Gamma) \ge -\sum_{\gamma \in \Gamma} a(\gamma)$*.*

With the incompatibility relation "supports meet" the hypothesis of Theorem 7.2 is literally the conclusion of Theorem 7.1. For the box gas the three steps are composed in one file:

Theorem 7.3 (Lean: `BoxGasFromFaceSum.box_dobrushin_of_face_sum`, `box_gas_partition_of_face_sum`)**.** *Let* $\mathcal{X}$ *be a finite family of plaquette sets,* z *real, and suppose that every box image* $\Gamma = \mathrm{boxImage}(X)$*,* $X \in \mathcal{X}$*, has* $|\operatorname{supp}\Gamma| \ge 1$*, and that (H-sum) holds in the box form,* $\sum_{X \in \mathcal{X},\ b \in \operatorname{supp}\mathrm{boxImage}(X)} |z(X)| e^{2|\operatorname{supp}\mathrm{boxImage}(X)|} \le 1$ *for every box* b*. Then the box gas (polymers* $\mathrm{boxImage}(\mathcal{X})$*, activity* z_B*, incompatibility "box supports meet") satisfies (D) with* $a(\Gamma) = \log(1 + |z_B(\Gamma)| e^{2|\operatorname{supp}\Gamma|})$*, and for every* $\Gamma_s \subseteq \mathrm{boxImage}(\mathcal{X})$*:* $Z(\Gamma_s) > 0$*,* $Z(\Gamma_s) \ge \exp(-\sum_{\Gamma \in \Gamma_s} a(\Gamma))$ *and* $\log Z(\Gamma_s) \ge -\sum_{\Gamma \in \Gamma_s} a(\Gamma)$*.*

```
theorem box_gas_partition_of_face_sum (ΛL : Finset Link) (m : ℕ) (𝒳 : Finset (Finset
      Face))
    (z : Finset Face → ℝ)
    (hsize : ∀ Γ ∈ 𝒳.image (boxImage ΛL m), 1 ≤ (boxSupp Γ).card)
    (hsum : ∀ b : Pt, ∑ X ∈ 𝒳.filter (fun X => b ∈ boxSupp (boxImage ΛL m X)),
      |z X| * Real.exp (2 * (boxSupp (boxImage ΛL m X)).card) ≤ 1) :
    ∀ Γs ⊆ 𝒳.image (boxImage ΛL m),
      0 < PolymerGasDobrushin.Z boxInc (zBox ΛL m 𝒳 z) Γs ∧
```

```
    Real.exp (-∑ Γ ∈ Γs, boxA ΛL m 𝒳 z Γ) ≤ PolymerGasDobrushin.Z boxInc (zBox ΛL m
    𝒳 z) Γs ∧
    -∑ Γ ∈ Γs, boxA ΛL m 𝒳 z Γ ≤ Real.log (PolymerGasDobrushin.Z boxInc (zBox ΛL m
    𝒳 z) Γs) := by
  have hcond := box_dobrushin_of_face_sum ΛL m 𝒳 z hsize hsum
```

The smallness hypothesis $|z_B(\Gamma)|e^{2|\operatorname{supp}\Gamma|} \le e-1$ of Theorem 7.1 is not assumed: by Theorem 6.1 the term of Γ in the box-stage sum at a box $b \in \operatorname{supp}\Gamma$ is at most $1 \le e-1$. The file `BoxGasFromFaceSum.lean` consists of verbatim copies of `BoxFromFace.lean` and `PolymerGasDobrushin.lean` (each in its own namespace) followed by the two composed theorems and two auxiliary lemmas (symmetry of the incompatibility, $a \ge 0$).

Tools for checking (H-sum). `ClassSiteSum.hsite_of_class_sum` derives (KP) from $|z(\gamma)| \le c\,\varepsilon_{\mathrm{cls}\gamma}^{|\operatorname{supp}\gamma|}$ and counts $N(k,n)$ of clusters through a point by class and size, when $c\sum_k \sum_{n\le M} N(k,n)\varepsilon_k^n e^{2n} \le 1$; `SizeSplitKP.size_split_site_sum` bounds the clusters above a size n_0 by a geometric series when the counts are $\le cK^{n-1}$ and $K\eta < 1$.

8 What is decided and what is proved

A statement is *decided* when its whole proof is `by decide +kernel`: the kernel evaluates the finite sets (built from `Finset` operations on `Fin`, $\mathbb{Z}$ and the tables `links0`, `faces0`) and checks the claim. All other statements have tactic proofs. Table 1 counts theorem declarations per file (one per `#print axioms` line; the two `@[simp]` lemmas `tF_apply`, `tL_apply`, proved by `rfl`, are not printed and not counted) and separates verbatim copies from new statements.

Table 1: Theorem declarations. "Copied": same name and statement as in the earlier file named; "decided": whole proof `by decide +kernel`.

File	theorems	copied	new	decided (new)
`ConnectedSetCounting`	14	–	14	–
`TwoColorPolymerSum`	18	3	15	–
`KPToDobrushin`	2	–	2	–
`TwoColorEncoding`	37	20	17	–
`LatticeTwoColorKP`	68	37	31	6
`BoxTwoColorKP`	71	51	20	5
`Lattice24KP`	73	53	20*	4
`BoxFromFace`	100	81	19	1
`PolymerGasDobrushin`	8	–	8	–
`ClassSiteSum`	4	–	4	–
`SizeSplitKP`	6	–	6	–
`BoxConnectivity`	5	–	5	–
`BoxGasFromFaceSum`	112	108	4	–
total	518	353	165	16

* Includes `quad_le_pow` and `exp_two_gt`, which also occur in `BoxTwoColorKP`; counted once among the 159 distinct statements. Copies: `TwoColorPolymerSum` §1 from `ConnectedSetCounting`; `TwoColorEncoding` §§1–6 from `TwoColorPolymerSum` and §10 from `KPToDobrushin`; `LatticeTwoColorKP` §§1–10 from `TwoColorEncoding` with the neighbourhood hypothesis weakened to $p \ne q$; `BoxTwoColorKP` and `Lattice24KP` §§1–11 from `LatticeTwoColorKP`; `BoxFromFace` §§1–12 from `BoxTwoColorKP`, and its §15 repeats the five decided degree counts of `LatticeTwoColorKP`; `BoxGasFromFaceSum` contains `BoxFromFace` and `PolymerGasDobrushin` whole.

Counting each decided fact once, 16 distinct statements are decided: the face degrees $20, 44, 66, 122$ and the four links of a plaquette; the 20 box shifts (twice, as `card_boxOff`

and `card_offFF`, built differently); $|\{-1,0,1\}^4| = 81$; the three coordinate bounds $|\cdot| \le 2$; $|\{0,1\}^4| = 16$; and the degrees $24, 44, 66, 122$ of the wider adjacency. One further statement, `box_diag`, uses `decide` on concrete memberships inside a short proof. The remaining 147 distinct statements, including every inequality on sums, all statements about connectedness and the encoding, and all arithmetic with e (by `norm_num` after $e \le 2.7182818286$ or $7.389 < e^2$), are proofs. A decided count says that a set *defined in Lean* has the stated size; that the definition matches the intended geometric neighbourhood is checked by the membership lemmas (`adj_mem_nbFF`, `fl_mem`, `lf_mem`, `ll_mem`, `box_offset`), which are proofs, in one direction only: every true neighbour lies in the set. Sets that are too large would only weaken the bounds.

9 What remains an assumption

Table 2: Status of each ingredient.

Ingredient	Grade
encoding: connectedness, recovery, size, label and fibre sums	Lean
two-colour KP sum (Lemma 3.1, Proposition 3.2)	Lean
degrees $20, 44, 66, 122$; 4 links; 20 box shifts; 16 points	Lean (`decide +kernel`)
arithmetic sufficient conditions (Lemma 3.3, `box_fix`, `lattice24_fix`)	Lean (`norm_num`)
(KP) $\Rightarrow$ (D) $\Rightarrow$ $Z > 0$, $\log Z \ge -\sum a$	Lean
(H-sum) $\Rightarrow$ (D) and Z bounds for the box gas, Theorem 7.3	Lean
identity of Theorem 6.1; bridge, Theorem 6.2	Lean
(H-2col) on the plaquette stage, label-sum bounds	assumption (hypotheses `hz`, `hlink`)
(H-sum)	assumption
product form over boxes (`box_hz_from_face`, `box_kp_site`)	assumption, not relied on
clusters form a gas, incompatibility = meeting supports	assumption, not formalised
Peierls-type bound $\mu(\text{box bad}) \le \sum_{X \cap b \ne \emptyset} \rho^*(X)$	by hand, not formalised
box degrees $39, 39, 56$; split $8 + 12$ of the shifts	computed outside Lean, unused

The product form over boxes asks that the activity of a box configuration be bounded by a product of per-box factors; the authors' numerical exploration indicates that correlations between plaquettes selected in different boxes make it unsuitable for the model they have in mind, so the identity of Theorem 6.1, with (H-sum) as the assumption, is the route kept here. Whether (H-sum) holds for a particular gauge model is not addressed.

10 What is machine-checked

The thirteen files of Table 1 each begin with `import Mathlib` and are checked as single files. They are distributed as one bundle (ledger and build instructions: `https://computoergosum.com/en/principia/lean/index.html`; bundle: `https://computoergosum.com/principia/lean/principia-ym-two-colour-polymer-2026-10-30.tar.gz`, checksum file `https://computoergosum.com/principia/lean/principia-ym-two-colour-polymer-2026-10-30.tar.gz.sha256`; axiom log: `https://computoergosum.com/principia/lean/axioms-ym-two-colour-polymer-2026-10-30.txt`). In the repository named in §13 the 13 files are kept, byte for byte the same, under `lean/ym-two-colour-polymer` as the modules `TwoColourPolymer.<name>` of a lake project pinned to the same versions; verify there with `lake exe cache get` and `LEAN_NUM_THREADS=1 lake env lean TwoColourPolymer/<name>.lean` for each file. To verify: extract the

bundle; place the files in a Lake project pinned to Lean v4.33.1 and Mathlib v4.33.1 (commit `0df444a3`); run `lake exe cache get`; then run `LEAN_NUM_THREADS=1 lake env lean <file>.lean` for each file and compare the `#print axioms` lines at its end with `axioms-<file>.txt`. There is no `sorry`, no `axiom` declaration and no `native_decide`. The file carrying the main statements of §6, `BoxFromFace.lean` (namespace `BoxFromFace`, 100 theorems), has sha256 `fbab4754f910061a4581d78c80dd35f05335e3c82abc0a296e6d027f12016f69`; `LatticeTwoColorKP.lean` (Theorem 5.1) has `0f24fff2e02976f50574d54931cef829a1314fc54e2983b7e0931396dd6fa049`; `BoxGasFromFaceSum.lean` (Theorem 7.3) has `f659c624ac9f43d2a5a2a5e4e751cb6399a10c3496781dcaf2258f78fbb12b3c`. The checksums of all thirteen files are in the file `SHA256SUMS` of the bundle.

Every theorem depends on at most `propext`, `Classical.choice` and `Quot.sound`; the only exception in form is `blk_add` (in each file that contains it), which depends on `propext` and `Quot.sound` only. In the logs of `BoxFromFace`, `BoxGasFromFaceSum`, `Lattice24KP`, `LatticeTwoColorKP`, `BoxTwoColorKP`, `TwoColorEncoding`, `TwoColorPolymerSum` and `ConnectedSetCounting` the axiom is printed as `choice` rather than `Classical.choice`: these files contain `open Classical`, under which `#print axioms` shows the short name of the same constant. It is not a different axiom.

Lean checks the rows marked "Lean" in Table 2, in the generality of the hypotheses quoted; it does not check that `hz`, `hlink` or (H-sum) hold for any model, nor the polymer form of a gauge model's large-field expansion.

11 What is not claimed

- No bound on the activities of any lattice gauge model is claimed or derived. The numerical constants in the hypotheses ($2.5 \cdot 10^{-3}$, $1.5 \cdot 10^{-11}$, $2.55 \cdot 10^{-3}$, 10^{-9}, $2.6 \cdot 10^{-5}$, $2.1 \cdot 10^{-3}$) are thresholds of sufficient conditions, not estimates.
- (H-2col), the label-sum bounds and (H-sum) are assumptions. We do not claim that they hold for the $SU(2)$ model that motivated this work; the product form over boxes is kept only as a formal statement.
- That the large-field expansion of a gauge theory yields a polymer gas of the kind of §2 (compatibility by disjoint supports, activities attached to clusters) is not formalised.
- The degrees used for pairs involving a defect on the box stage are upper bounds $(81, 81, 80)$, not exact counts; the degrees 48, 72 in Lemma 3.3 are upper bounds for the decided 44, 66.
- The cluster-counting bounds are of standard type [3, 1, 5]; we do not know whether the particular two-colour encoding with labelled defect links has been noted before.

12 Correspondence table

Table 3: Statements of this note and Lean declarations. Where a declaration is copied into later files, the line of the file named is given.

Statement	Lean declaration	File:line
Lemma 3.1	`Wg_le`	LTK:148
count $\le e^n d^{n-1}$	`connected_count_le`	CSC:249
one-colour KP sum $\le 1/d$	`point_kp_sum_le`	CSC:420
Proposition 3.2, generating function	`two_color_Wg_le`	LTK:243

Statement	Lean declaration	File:line
Proposition 3.2, point	`two_color_point_sum`	LTK:274
Proposition 3.2, root set	`two_color_root_set_sum`	LTK:311
Proposition 3.2, point weights	`two_color_root_gf`	LTK:833
Lemma 3.3	`lattice_fix`	LTK:397
Lemma 3.3, $\theta_0 + 4\theta_1 \le 1$	`lattice_face_total`	LTK:427
abstract face KP sum (first version)	`lattice_two_color_kp`	TPS:435
def, thin, enc, `cadj`	`defects`, `thin`, `enc`, `cadj`	LTK:539
Lemma 4.1(i)	`enc_connected`	LTK:582
Lemma 4.1(ii)	`decode`, `enc_eq`, `thin_disjoint`	LTK:629
Lemma 4.1(iii)	`card_le_enc`	LTK:663
Lemma 4.1(iv)	`label_sum_le`	LTK:490
Proposition 4.2	`encoded_sum_le`	LTK:715
Proposition 4.2, weight $e^{c\|X\|}$	`encoded_sum_le_c`	BFF:1523
abstract (KP), plaquette stage	`encoded_kp_site`	LTK:876
abstract (D), plaquette stage	`encoded_dobrushin`	LTK:1030
$f \in D_\ell \iff \ell \in \partial f$	`mem_facesOf_iff`	LTK:1107
plaquette adjacency	`faceAdj`	LTK:1111
membership in $N_{FF}, N_{FL}, N_{LF}, N_{LL}$	`adj_mem_nbFF`, `fl_mem`, `lf_mem`, `ll_mem`	LTK:1116
degrees $20, 44, 66, 122$; 4 links (decided)	`card_nbFF0`, `card_nbFL0`, `card_nbLF0`, `card_nbLL0`, `card_links0`	LTK:1171
`hnb` on $\mathbb{Z}^4$	`lattice_hnb`	LTK:1187
`hdeg` on $\mathbb{Z}^4$	`lattice_hdeg`	LTK:1211
`hfour` on $\mathbb{Z}^4$	`lattice_hfour`	LTK:1220
Theorem 5.1, (KP)	`lattice_kp_site`	LTK:1228
Theorem 5.1, (D)	`lattice_dobrushin`	LTK:1240
shift of adjacent plaquettes	`box_offset`	LTK:1280
20 box shifts (decided)	`card_boxOff`	LTK:1291
box degree ≤ 20	`box_degree`	LTK:1299
box image connected	`box_image_connected`	LTK:1310
diagonal shift $(1, -1, 0, 0)$	`box_diag`	LTK:1328
wider adjacency	`cube0pp0`, `faceAdj24`	L24:1172
degrees $24, 44, 66, 122$ (decided)	`card_nbFF24_0` … `card_nbLL24_0`	L24:1237
wider adjacency, arithmetic	`lattice24_fix`	L24:1300
wider adjacency, (KP)	`lattice_kp_site24`	L24:1390
wider adjacency, sharpness	`face24_fails`	L24:1432
20 thin box shifts (decided)	`card_offFF`	BTK:1207
$\|\{-1, 0, 1\}^4\| = 81$ (decided)	`card_cube81`	BTK:1212
coordinate bounds (decided)	`nbFL0_bdd`, `nbLF0_bdd`, `nbLL0_bdd`	BTK:1215
box degrees $(20, 81, 81, 80)$	`card_offc`	BTK:1243
box `hnb`, `hdeg`	`box_hnb`, `box_hdeg`	BTK:1258
box image of a two-colour set	`bimg_connected`	BTK:1310
box arithmetic	`box_fix`	BTK:1334
box (KP), root box	`box_kp_site`	BTK:1361
box sharpness	`box_fails`	BTK:1402
box (KP), supp = boxes	`box_hsite`	BTK:1432
box (D)	`box_dobrushin`	BTK:1464
16 points (decided)	`card_corners`	BFF:1662
fibre over a box set	`fiber_box_le`	BFF:1610
product form from (H-2col)	`box_hz_from_face`	BFF:1744
ε_{box} arithmetic	`epsBox_numbers`	BFF:1785
plaquette stage $\Rightarrow$ box (D)	`face_to_box_dobrushin`	BFF:1842
Theorem 6.1, identity	`box_hsite_eq_face_sum`	BFF:1807
Theorem 6.1, (H-sum) $\Rightarrow$ box (KP)	`box_hsite_of_face_sum`	BFF:1833
root set of a box, ≤ 384 links	`boxRoot`, `card_boxRoot_links`	BFF:1940
Theorem 6.2	`box_event_bridge`	BFF:1970
arithmetic instances (inputs not discussed)	`bridge_numbers_gamma`, `bridge_numbers_chernoff`	BFF:2009
sum over meeting polymers	`intersect_sum_le`	KPD:27
Theorem 7.1	`dobrushin_cond_of_kp`	KPD:46

Statement	Lean declaration	File:line
Theorem 7.2, $Z > 0$	`dobrushin`	PGD:118
Theorem 7.2, lower bounds	`Z_ge_exp`, `log_Z_ge`	PGD:194
class sum $\Rightarrow$ (KP)	`hsite_of_class_sum`	CSS:74
size split	`size_split_site_sum`	SSK:80
Theorem 7.3, (D) for the box gas	`box_dobrushin_of_face_sum`	BGF:2396
Theorem 7.3, Z bounds	`box_gas_partition_of_face_sum`	BGF:2428
incompatibility, $a \geq 0$ (auxiliary)	`boxInc_symm`, `boxA_nonneg`	BGF:2384

File abbreviations: CSC `ConnectedSetCounting`, TPS `TwoColorPolymerSum`, TCE `TwoColorEncoding`, LTK `LatticeTwoColorKP`, BTK `BoxTwoColorKP`, L24 `Lattice24KP`, BFF `BoxFromFace`, KPD `KPToDobrushin`, PGD `PolymerGasDobrushin`, CSS `ClassSiteSum`, SSK `SizeSplitKP`, BCN `BoxConnectivity`, BGF `BoxGasFromFaceSum`. Namespaces equal file names. BCN (blocks of side L are boxes; a box with points in and out of A has an edge leaving A) is in the bundle but not used above.

13 Disclosure of AI use

This note was produced with the assistance of AI agents. The encoding, the Lean 4 files including the formalised statements, and the reports on which this note is based were written by instances of Claude (Anthropic) run as the coding agent Claude Code on a laptop, across many separate sessions, each of which began without memory of the previous ones and read the written record instead. A second group of instances, running on the server of the website named on the title page, is responsible for re-running the Lean files, for checking the statements of this note against the files and the reports, and for preparing the distributed version; the text of this note was prepared by that group from the files and reports. The human author set the task, chose the problem, fixed the working rules and load limits, and decides whether and where the result is reported; he is responsible for this note. Adversarial review was carried out only by further instances of the same model family, which share the same blind spots. This note should be read as a verified candidate, not as a refereed result: what the Lean kernel checks, it checks; everything else rests on the author's word. This edition and the Lean files it cites are kept in the repository `https://github.com/yoshikazuhosomura-sys/principia` (the note under `papers/ym-two-colour-polymer`, the Lean files under `lean/ym-two-colour-polymer`). The web edition of this note, published on the site named on the title page, lists the same work under the names Kiichi, Shiori and Rin, the human author and the two groups of AI agents; the two editions differ only in the title page, in this section and in one sentence of §10 giving the location of the Lean files in the repository.

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